

Ευφυείς Πράκτορες και Πολυπρακτορικά Συστήματα

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Ευφυείς Πράκτορες και Πολυπρακτορικά Συστήματα

- Θεμελιώδεις έννοιες σε ότι αφορά τους ευφυείς πράκτορες, και τα πολυπρακτορικά συστήματα.
- σχεδιασμός αλγορίθμων που αφορούν στην κατανεμημένη επίλυση προβλημάτων, στην επίτευξη στόχων μέσω ανταγωνιστικότητας ή συντονισμού και συνεργασίας.

Ευφυείς Πράκτορες και Πολυπρακτορικά Συστήματα

- Σχεδιασμός αλγορίθμων που αφορούν στην κατανεμημένη μάθηση πολλαπλών πρακτόρων.
- Επικοινωνία ιδεών για πράκτορες και πολυπρακτορικά συστήματα με καθαρό, σαφή και τυπικό τρόπο.

Ευφυείς Πράκτορες και Πολυπρακτορικά Συστήματα

- Εισαγωγικά στοιχεία θεωρίας για τους ευφυείς πράκτορες και τη θεωρία χρησιμότητων
- Βασικές αρχιτεκτονικές ανάπτυξης πρακτόρων.
- Πολυπρακτορικές αλληλεπιδράσεις: Συνεργασία, συντονισμός. Βασικά θέματα θεωρίας παιγνίων.
- Θεωρία και αλγοριθμικά θέματα μη συνεργατικών παιγνίων.
- Κατανεμημένη επίλυση προβλημάτων: Κατανομή προβλημάτων και συντονισμός για την επίλυση προβλημάτων.
- Κατανεμημένη επίλυση προβλημάτων περιορισμών, κατανομή εργασιών σε πράκτορες, κατανεμημένος δυναμικός προγραμματισμός, κοινωνικοί κανόνες.
- Σύναψη συμφωνιών: Σχεδιασμός μηχανισμού, δημοπρασίες, διαπραγμάτευση.
- Επικοινωνία μεταξύ πρακτόρων.
- Λήψη Σύνθετων αποφάσεων – επίλυση ακολουθιακών προβλημάτων

Ευφυείς Πράκτορες και Πολυπρακτορικά Συστήματα

Εξέταση μαθήματος

- Εργασία με παράδοση αναφοράς
- Παρουσίαση της εργασίας – προφορική εξέταση

Ευφυείς Πράκτορες και Πολυπρακτορικά Συστήματα

Βασική βιβλιογραφία

- Michael Wooldridge, Introduction to MultiAgent Systems (Ελληνική έκδοση «Εισαγωγή στα Πολυπρακτορικά συστήματα», εκδόσεις Κλειδάριθμος), 2008.
- Yoav Shoham, Kevin Leyton-Brown Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations, Cambridge University Press, 2009

Agents

$$act = \text{AgentFn}(percept)$$

sensors

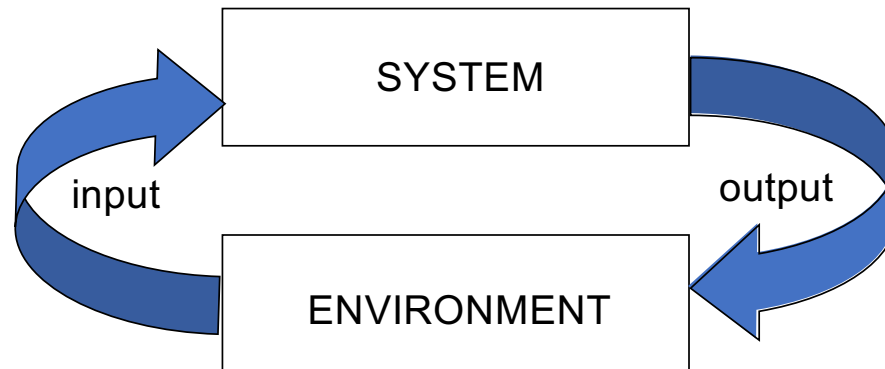
agent fn

actuators



What is an Agent?

- The main point about agents is they are *autonomous*: capable of acting independently, exhibiting control over their internal state
- Thus: *an agent is a computer system capable of autonomous and flexible action in some environment in order to meet its design objectives*



Abstract Architecture for Agents

- Assume the environment may be in any of a finite set E of discrete, instantaneous states:

$$E = \{e, e', \dots\}.$$

- Agents are assumed to have a repertoire of possible actions available to them, which transform the state of the environment:

$$Ac = \{\alpha, \alpha', \dots\}$$

- A *run*, r , of an agent in an environment is a sequence of interleaved environment states and actions:

$$r : e_0 \xrightarrow{\alpha_0} e_1 \xrightarrow{\alpha_1} e_2 \xrightarrow{\alpha_2} e_3 \xrightarrow{\alpha_3} \dots \xrightarrow{\alpha_{u-1}} e_u$$

Abstract Architecture for Agents

- Let:
 - R be the set of all such possible finite sequences (over E and Ac)
 - R^{Ac} be the subset of these that end with an action
 - R^E be the subset of these that end with an environment state

State Transformer Functions

- A *state transformer* function represents behavior of the environment:

$$\tau : \mathcal{R}^{Ac} \rightarrow \wp(E)$$

- Note that environments are...
 - *history dependent*
 - *non-deterministic*
- If $\tau(r) = \emptyset$, then there are no possible successor states to r . In this case, we say that the system has *ended* its run
- Formally, we say an environment Env is a triple $Env = \langle E, e_0, \tau \rangle$ where: E is a set of environment states, $e_0 \in E$ is the initial state, and τ is a state transformer function

Agents

- Agent is a function which maps runs to actions:

$$Ag : \mathcal{R}^E \rightarrow Ac$$

An agent makes a decision about what action to perform based on the history of the system that it has witnessed to date. Let AG be the set of all agents

Systems

- A *system* is a pair containing an agent and an environment
- Any system will have associated with it a set of possible runs; we denote the set of runs of agent Ag in environment Env by $R(Ag, Env)$
- (We assume $R(Ag, Env)$ contains only *terminated* runs)

Systems

- Formally, a sequence

$$(e_0, \alpha_0, e_1, \alpha_1, e_2, \dots)$$

represents a run of an agent Ag in environment $Env = \langle E, e_0, \tau \rangle$ if:

1. e_0 is the initial state of Env
2. $\alpha_0 = Ag(e_0)$; and
3. For $u > 0$,

$$\begin{aligned} e_u &\in \tau((e_0, \alpha_0, \dots, \alpha_{u-1})) \quad \text{where} \\ \alpha_u &= Ag((e_0, \alpha_0, \dots, e_u)) \end{aligned}$$

Agent Control Loop

1. Agent starts in some initial internal state i_0
2. Observes its environment state e , and generates a percept $see(e)$
3. Internal state of the agent is then updated via $next$ function, becoming $next(i_0, see(e))$
4. The action selected by the agent is $action(next(i_0, see(e)))$
5. Goto 2

Utility Functions over States

- One possibility: associate *utilities* with individual states — the task of the agent is then to bring about states that maximize utility
- A task specification is a function

$$u : E \rightarrow \mathbb{R}$$

which associates a real number with every environment state

Utility Functions over States

- But what is the value of a *run*...
 - minimum utility of state on run?
 - maximum utility of state on run?
 - sum of utilities of states on run?
 - average?
- Disadvantage: difficult to specify a *long term* view when assigning utilities to individual states
(One possibility: a *discount* for states later on.)

Utilities over Runs

- Another possibility: assigns a utility not to individual states, but to runs themselves:

$$u : R \rightarrow \mathbb{R}$$

- Such an approach takes an inherently *long term* view
- Other variations: incorporate probabilities of different states emerging
- Difficulties with utility-based approaches:
 - where do the numbers come from?
 - we don't think in terms of utilities!
 - hard to formulate tasks in these terms

Utility in the Tileworld

- Simulated two dimensional grid environment on which there are agents, tiles, obstacles, and holes
- An agent can move in four directions, up, down, left, or right, and if it is located next to a tile, it can push it
- Holes have to be filled up with tiles by the agent. An agent scores points by filling holes with tiles, with the aim being to fill as many holes as possible
- TILEWORLD changes with the random appearance and disappearance of holes
- Utility function defined as follows:

$$u(r) \hat{=} \frac{\text{number of holes filled in } r}{\text{number of holes that appeared in } r}$$

The Tileworld, Some Examples

- From Goldman and Rosenschein, AAAI-94:

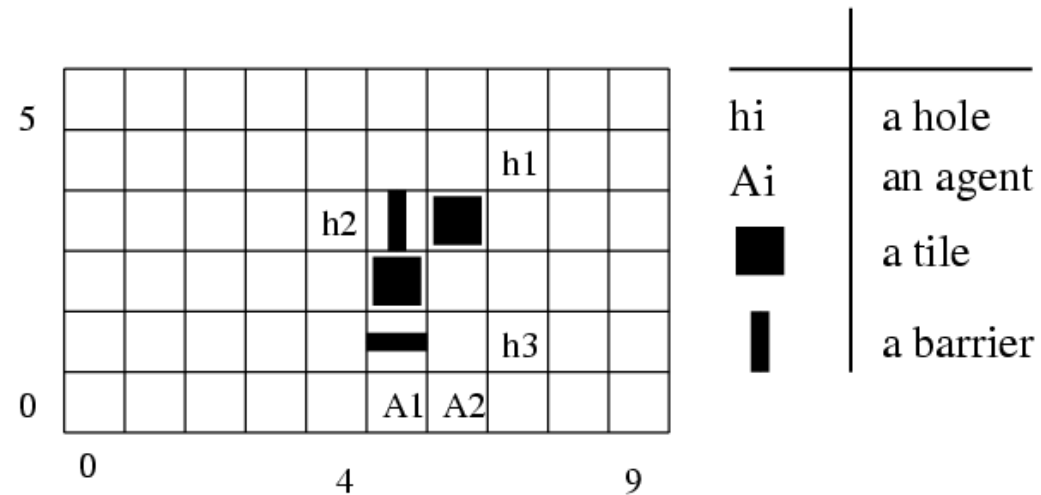


Figure 1: Strongly-Coupled Interactions

The Tileworld, Some Examples

- From Goldman and Rosenschein, AAAI-94:

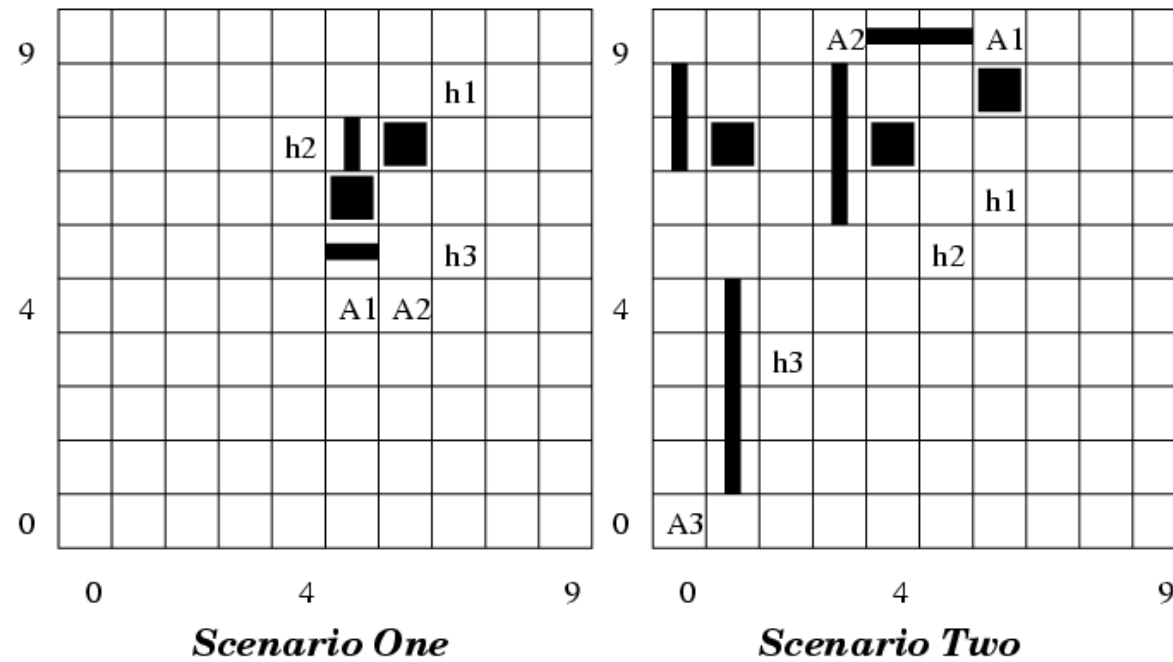


Figure 2: Simulations

Expected Utility & Optimal Agents

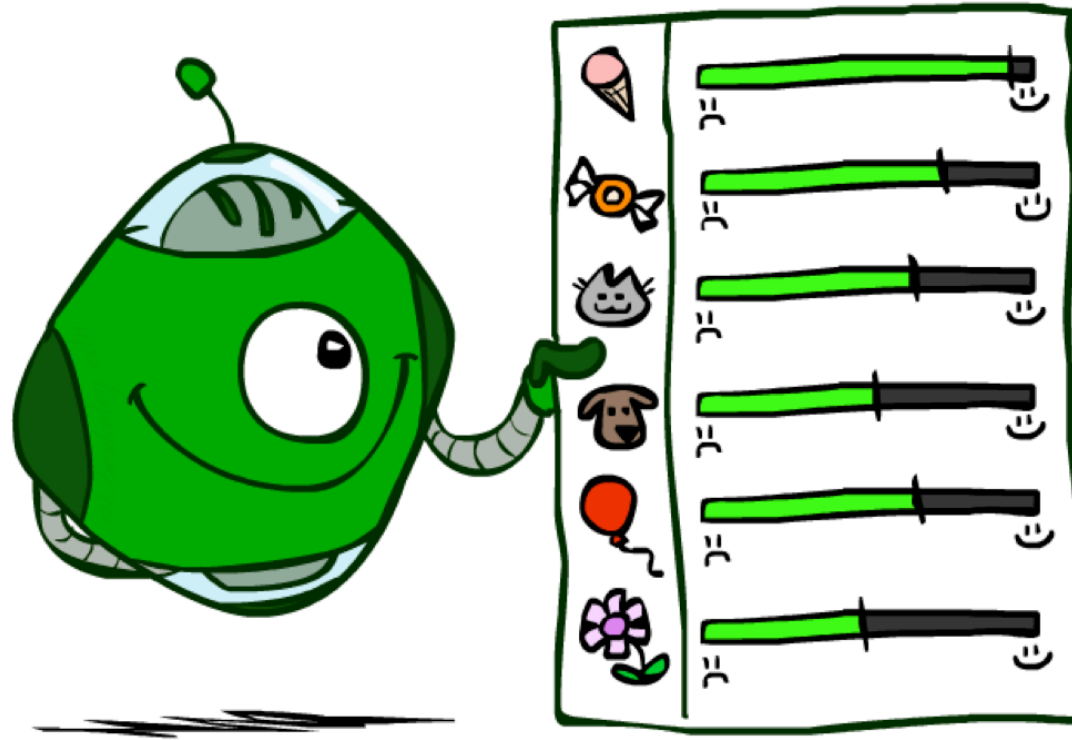
- Write $P(r \mid Ag, Env)$ to denote probability that run r occurs when agent Ag is placed in environment Env
Note:

$$\sum_{r \in \mathcal{R}(Ag, Env)} P(r \mid Ag, Env) = 1.$$

- Then optimal agent Ag_{opt} in an environment Env is the one that *maximizes expected utility*:

$$Ag_{opt} = \arg \max_{Ag \in \mathcal{AG}} \sum_{r \in \mathcal{R}(Ag, Env)} u(r)P(r \mid Ag, Env). \quad (1)$$

Utilities



Maximum Expected Utility

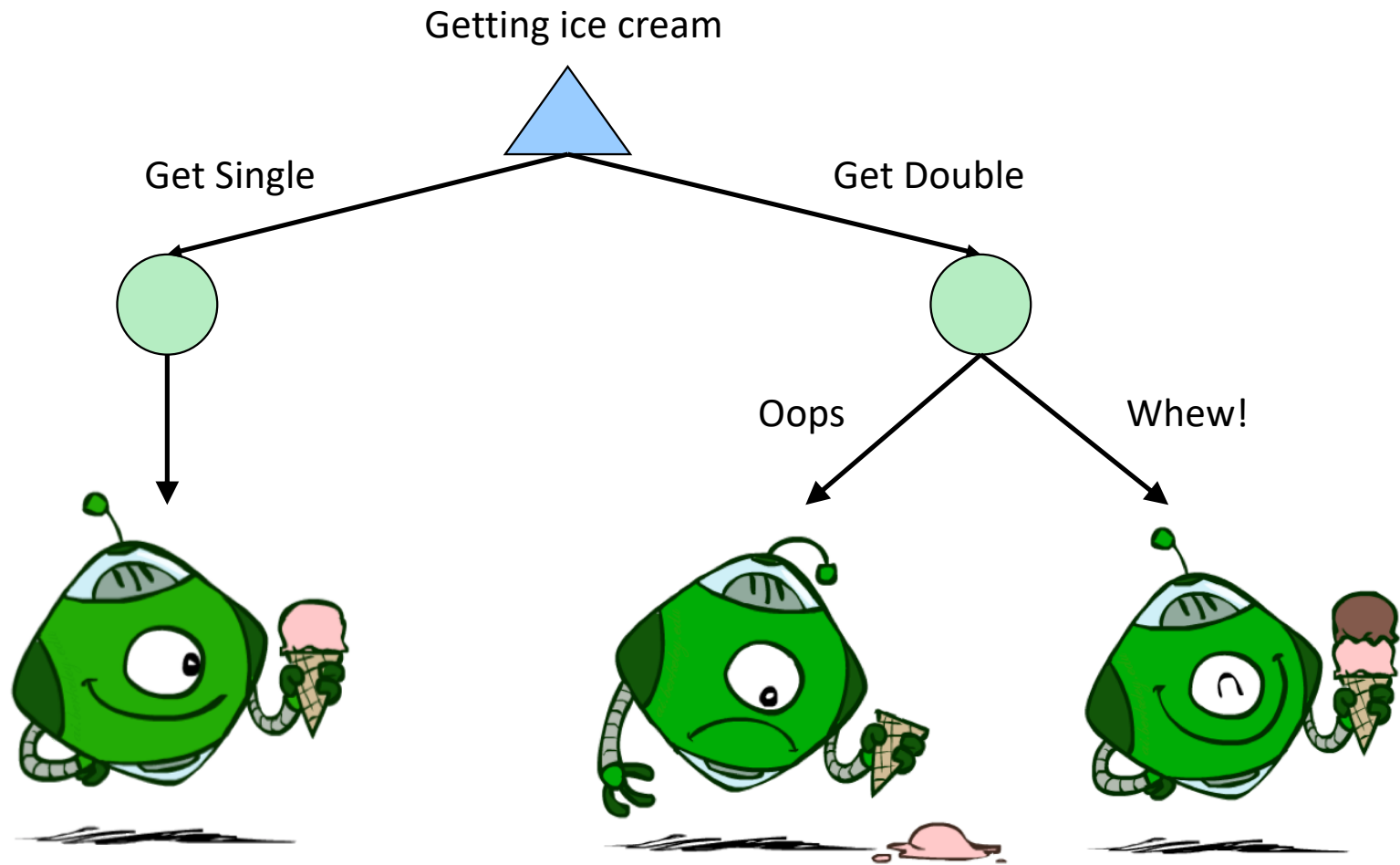
- Principle of maximum expected utility:
 - A rational agent should choose the action that **maximizes its expected utility, given its knowledge**
- Questions:
 - Where do utilities come from?
 - How do we know such utilities even exist?
 - How do we know that averaging even makes sense?
 - What if our behavior (preferences) can't be described by utilities?

Utilities

- Utilities are functions from outcomes (states of the world) to real numbers that describe an agent's preferences
- Where do utilities come from?
 - In a game, may be simple (+1/-1)
 - Utilities summarize the agent's goals
 - **Theorem:** *any "rational" preferences can be summarized as a utility function*
- We hard-wire utilities and let behaviors emerge
 - Why don't we let agents pick utilities?
 - Why don't we prescribe behaviors?



Utilities: Uncertain Outcomes



Maximizing the Expected Utility

- Expected Utility
 - $EU(A|E) = \sum_i P(Result_i(A) | Do(A), E) \cdot U(Result_i(A))$
- Computing probabilities: NP-Hard
- Computing utilities: Needs exploration.

Preferences

- An agent must have preferences among:

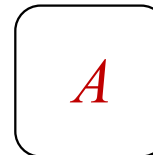
- Prizes: A , B , etc.
- Lotteries: situations with uncertain prizes

$$L = [p, A; (1 - p), B]$$

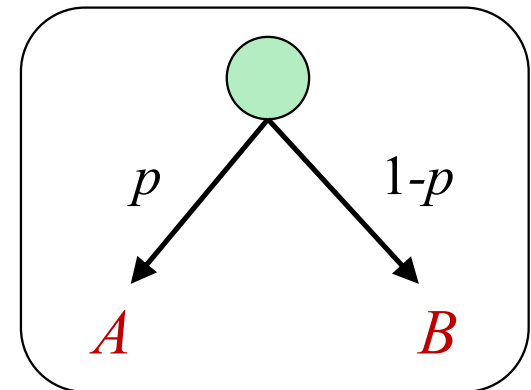
- Notation:

- Preference: $A \succ B$
- Indifference: $A \sim B$

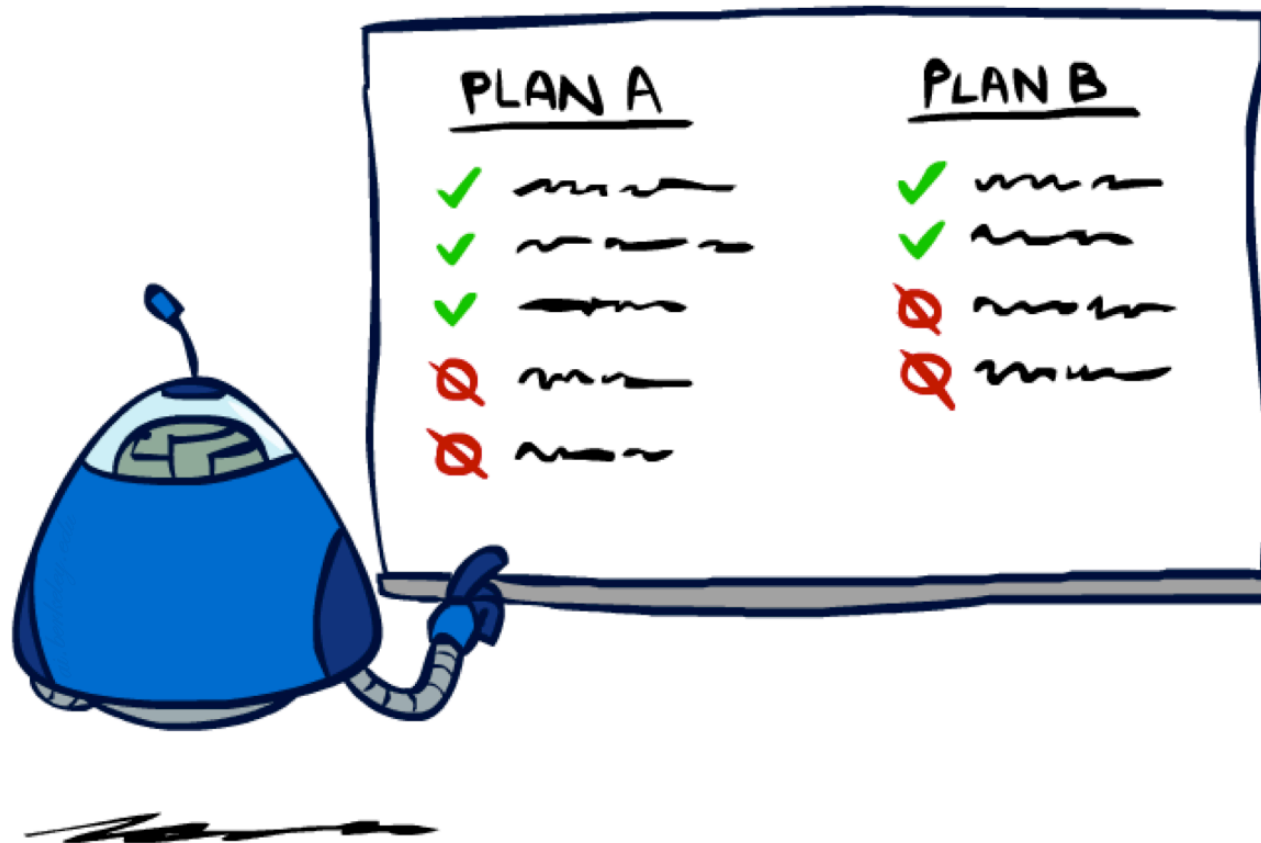
A Prize



A Lottery



Rationality

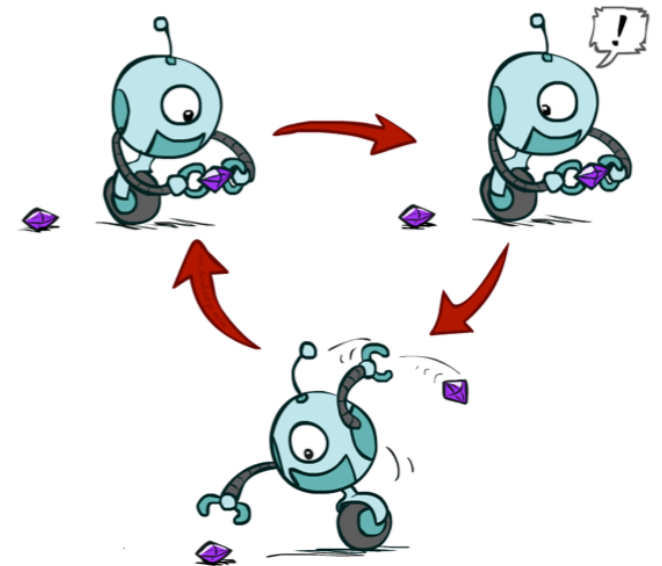


Rational Preferences

- We want some constraints on preferences before we call them rational, such as:

Axiom of Transitivity: $(A \succ B) \wedge (B \succ C) \Rightarrow (A \succ C)$

- For example: an agent with **intransitive preferences** can be induced to give away all of its money
 - If $B \succ C$, then an agent with C would pay (say) 1 cent to get B
 - If $A \succ B$, then an agent with B would pay (say) 1 cent to get A
 - If $C \succ A$, then an agent with A would pay (say) 1 cent to get C



Rational Preferences

The Axioms of Rationality

Orderability

$$(A \succ B) \vee (B \succ A) \vee (A \sim B)$$

Transitivity

$$(A \succ B) \wedge (B \succ C) \Rightarrow (A \succ C)$$

Continuity

$$A \succ B \succ C \Rightarrow \exists p [p, A; 1 - p, C] \sim B$$

Substitutability

$$A \sim B \Rightarrow [p, A; 1 - p, C] \sim [p, B; 1 - p, C]$$

Monotonicity

$$A \succ B \Rightarrow \\ (p \geq q \Leftrightarrow [p, A; 1 - p, B] \succeq [q, A; 1 - q, B])$$



Rational Preferences

- decomposability
 - $[p, A; 1 - p, [q, B; 1 - q, C]] \sim [p, A; (1 - p)q, B; (1 - p)(1 - q), C]$



Theorem: Rational preferences imply behavior describable as maximization of expected utility

MEU Principle

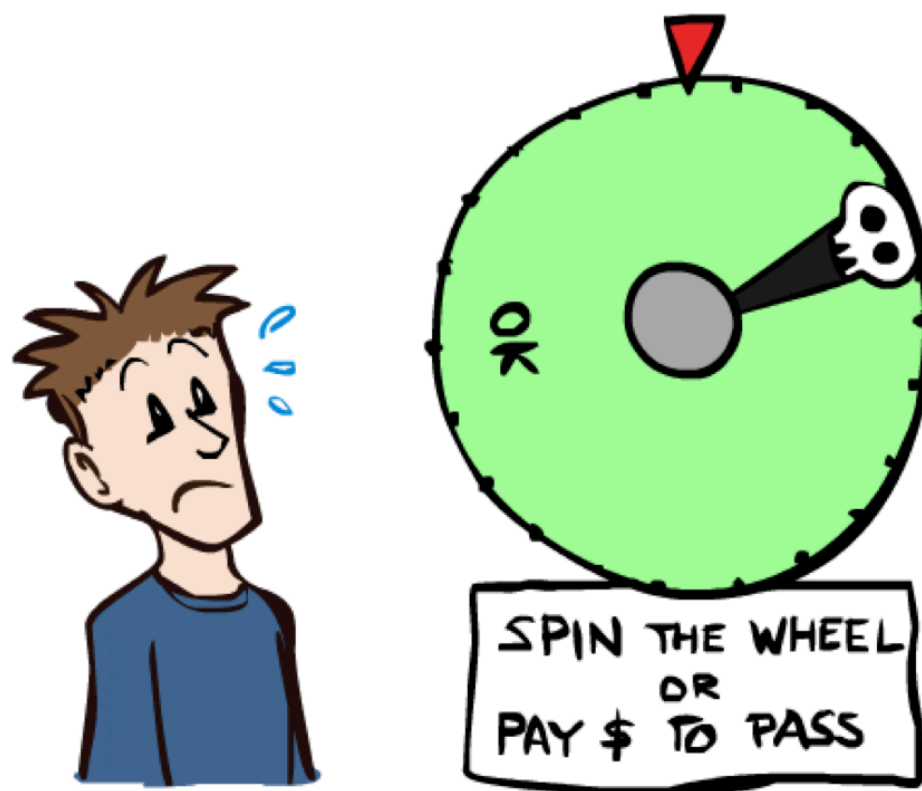
- Theorem [Ramsey, 1931; von Neumann & Morgenstern, 1944]
 - Given any preferences satisfying these constraints, there exists a real-valued function U such that:

$$U(A) \geq U(B) \Leftrightarrow A \succeq B$$

$$U([p_1, S_1; \dots ; p_n, S_n]) = \sum_i p_i U(S_i)$$

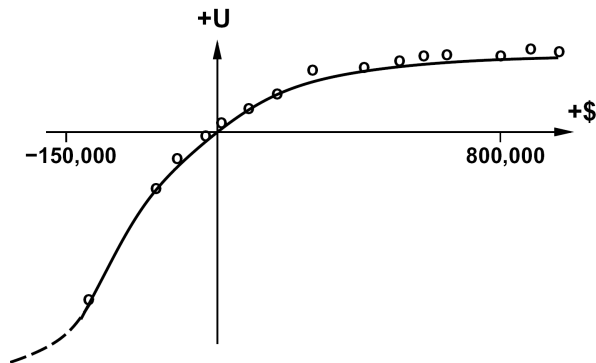
- I.e. values assigned by U preserve preferences of both prizes and lotteries!
- Maximum expected utility (MEU) principle:
 - Choose the action that maximizes expected utility
 - Note: an agent can be entirely rational (consistent with MEU) without ever representing or manipulating utilities and probabilities
 - E.g., a lookup table for perfect tic-tac-toe, a reflex vacuum cleaner

Human Utilities



Money

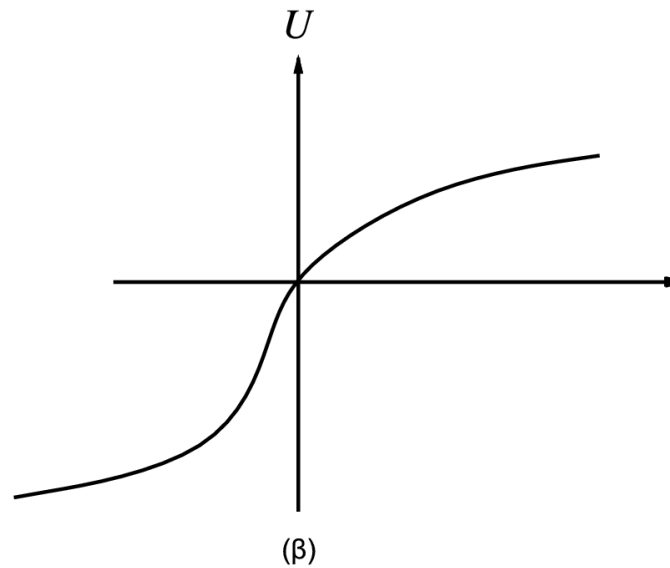
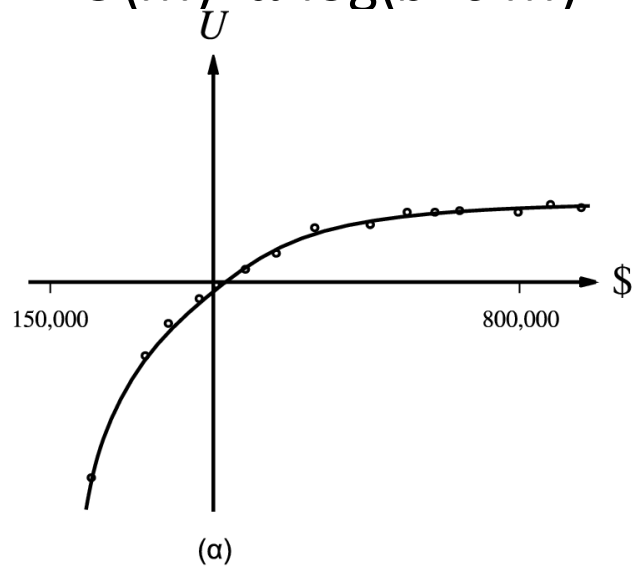
- Money does not behave as a utility function, but we can talk about the utility of having money (or being in debt)
- Given a lottery $L = [p, \$X; (1-p), \$Y]$
 - The **expected monetary value** $EMV(L)$ is $p \cdot X + (1-p) \cdot Y$
 - $U(L) = p \cdot U(\$X) + (1-p) \cdot U(\$Y)$
 - Typically, $U(L) < U(EMV(L))$
 - In this sense, people are **risk-averse**
 - When deep in debt, people are **risk-prone**



Utility of Money

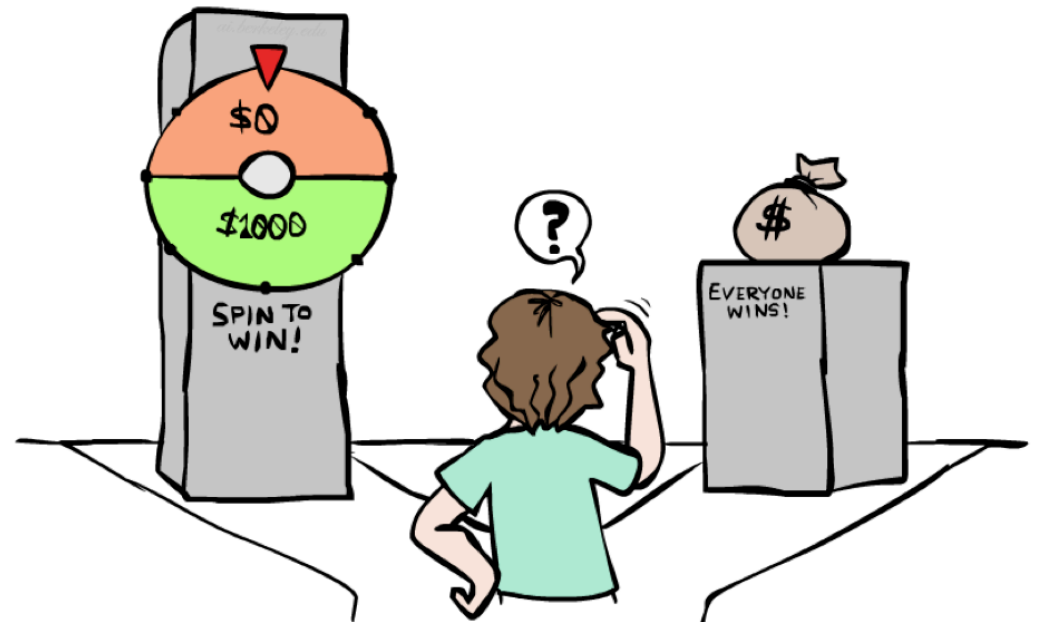
- A possible approach is by taking the log (Grayson 1960):

$$U(M) = \alpha \cdot \log(b + c \cdot M)$$



Example: Insurance

- Consider the lottery $[0.5, \$1000; 0.5, \$0]$
 - What is its **expected monetary value**? (\$500)
 - What is its **certainty equivalent**?
 - Monetary value acceptable in lieu of lottery
 - \$400 for most people
 - Difference of \$100 is the **insurance premium**
 - There's an insurance industry because people will pay to reduce their risk
 - If everyone were risk-neutral, no insurance needed!
 - It's win-win: you'd rather have the \$400 and the insurance company would rather have the lottery (their utility curve is flat and they have many lotteries)



Utility Scales

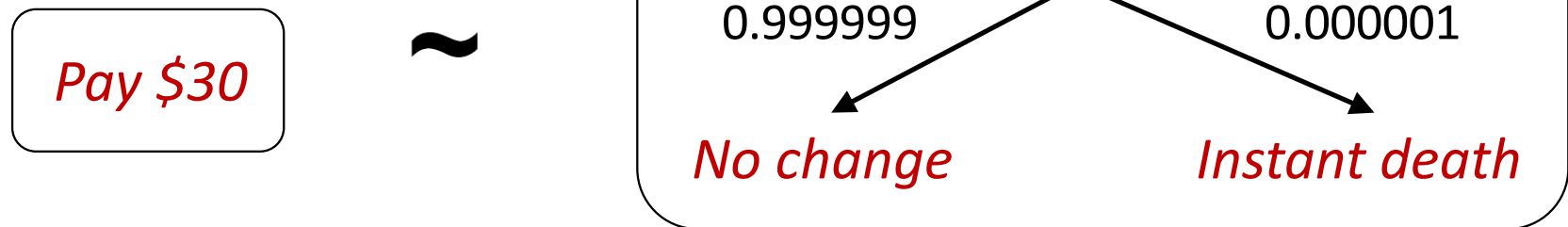
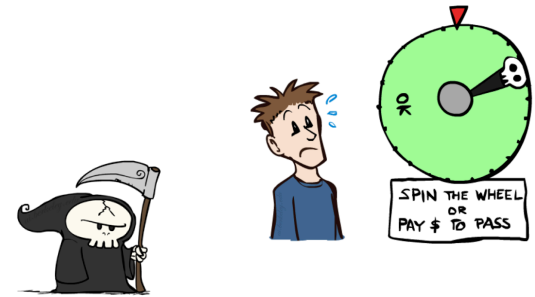
- Note: behavior is invariant under positive linear transformation

$$U'(x) = k_1 U(x) + k_2 \quad \text{where } k_1 > 0$$

- With deterministic prizes only (no lottery choices), only **ordinal utility** can be determined, i.e., total order on prizes
- **Normalized utilities**: $u_+ = 1.0$, $u_- = 0.0$
- Standard approach to assessment (elicitation) of human utilities:
 - Compare a prize A to a **standard lottery** L_p between
 - “best possible prize” u_+ with probability p
 - “worst possible catastrophe” u_- with probability $1-p$
 - Adjust lottery probability p until indifference: $A \sim L_p$
 - Resulting p

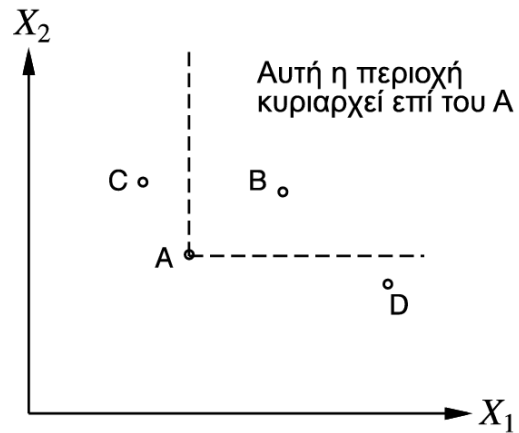
Human Utilities

- **Micromorts**: one-millionth chance of death, useful for paying to reduce product risks, etc.
- **QALYs**: quality-adjusted life years, useful for medical decisions involving substantial risk
 - is a utility in $[0,1]$

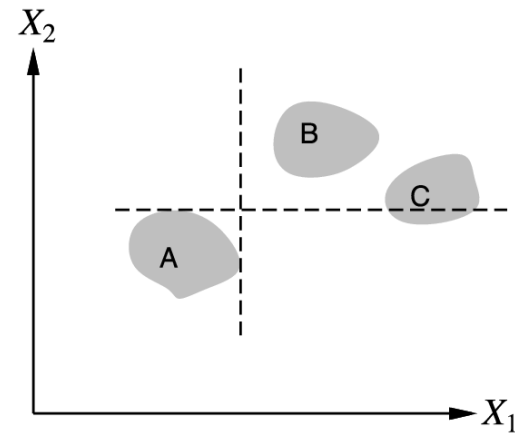


Multiattribute Utility Theory

- Attributes vector: $\mathbf{X} = \langle X_1, \dots, X_n \rangle$
- **Strict dominance** (deterministic and non-deterministic cases)



(α)



(β)

Multiattribute Utility Theory

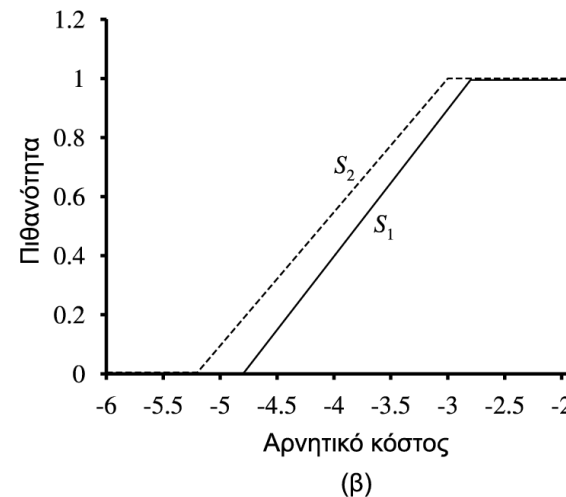
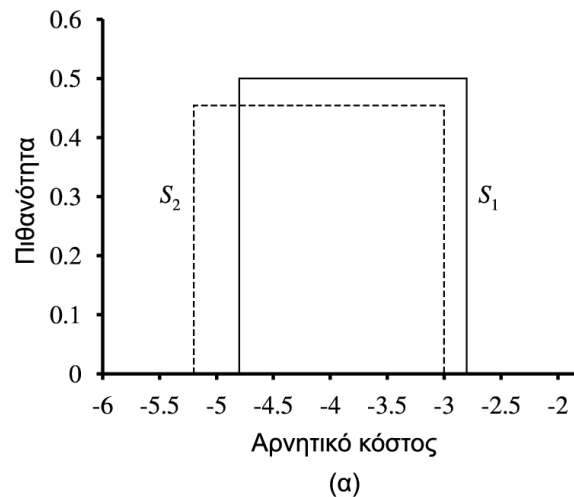
Stochastic dominance

- Cost of construction:

- S_1 : $U[2,8 \cdot 10^6 \text{ €}, 4,8 \cdot 10^6 \text{ €}](x)$
- S_2 : $U[3,0 \cdot 10^6 \text{ €}, 5,2 \cdot 10^6 \text{ €}](x)$
- Η χρησιμότητα φθίνει με το κόστος

Cumulative distributions:

$$\forall x \int_{-\infty}^x p_1(x') dx' \leq \int_{-\infty}^x p_2(x') dx'$$



Utilities Structure – deterministic case

- Representation theorem:

- $U(x_1, \dots, x_n) = f [f_1(x_1), \dots, f_n(x_n)]$

where f is a very simple function (usually addition).

- Preference independence of (X_1, X_2) w.r.t X_3 :

- (x_1, x_2, x_3) and (x'_1, x'_2, x_3) : The preference is independent to x_3 .

- Mutual Preference Independence (additive value function) – Debreu 1960:

$$V(x_1, \dots, x_n) = \sum_i V_i(x_i)$$

Utilities Structure – non-deterministic case

- Utility independence:
 - An attributes set **X** is independent w.r.t. utility from another attribute set **Y** if the preferences among the lotteries for attributes in **X** are independent from the values that attributes in **Y** get.
- Mutually utility-independence (MUI)
- Multiplicative utility function (3 attributes case):

$$U = k_1U_1 + k_2U_2 + k_3U_3 + k_1k_2U_1U_2 + k_2k_3U_2U_3 + k_3k_1U_3U_1 + k_1k_2k_3U_1U_2U_3$$

